Synthetic Tabular Data for Fraud Detection: CTGAN vs. FinDiff under Class-Ratio Control

Riye Kim
University of Birmingham
rxk355@student.bham.ac.uk

Abstract

Class imbalance remains a fundamental challenge in financial fraud detection, where fraudulent transactions represent only a small fraction of the data. Most existing studies evaluate synthetic tabular data through statistical similarity rather than downstream detection performance, and the effects of explicitly manipulating class ratios during generation remain poorly understood.

This study investigates how forced minority-class ratios influence generative quality and downstream fraud detection performance, comparing the GAN-based CTGAN [1] and the diffusion-based FinDiff [2] on the UCI Credit Card Fraud dataset using a fixed XGBoost classifier. Primary metrics are Fraud F1 and AUPRC; a generalisation study on the Taiwan Default Payment dataset incorporates SMOTE and TVAE as additional baselines.

The results reveal a clear asymmetry between the two paradigms. FinDiff is highly sensitive to ratio forcing, exhibiting monotonic distributional distortion and degraded detection performance as the forced minority ratio increases. CTGAN does not show the same degradation, but further analysis reveals that its conditional sampling mechanism frequently fails to enforce the requested ratios, inadvertently preserving the natural class distribution. A double-penalisation effect is also identified: combining synthetic ratio forcing with classifier-level reweighting causes catastrophic performance collapse. These findings demonstrate that explicit class-ratio manipulation is neither trivial nor consistently reliable across generative architectures.

1 Introduction

Financial fraud detection research is constrained by the sensitivity of real-world transaction data, which limits data sharing and reproducibility. Synthetic tabular data has therefore emerged as a promising alternative, enabling researchers to generate artificial datasets that resemble financial data without exposing sensitive information. However, most prior work evaluates synthetic data primarily through statistical similarity or sample realism, rather than its practical usefulness for downstream fraud detection tasks.

Another major challenge in fraud detection is extreme class imbalance. In real financial datasets, fraudulent transactions typically account for less than 1% of all observations. While synthetic data generation could potentially mitigate this imbalance by adjusting class proportions during sampling, it remains unclear how different generative architectures behave when such class-ratio manipulation is explicitly applied.

This study investigates how forcing minority-class ratios during synthetic tabular data generation affects both data quality and downstream fraud detection performance. Two representative generative paradigms are compared: the GAN-based generator CTGAN [1] and the diffusion-based tabular generator FinDiff [2]. The evaluation focuses on downstream detection performance, specifically Fraud F1 and AUPRC, as well as distribution fidelity and privacy-related characteristics of the generated datasets. Following Saito and Rehmsmeier [3], AUPRC and Fraud F1 are treated as primary metrics, as ROC-AUC can remain inflated even when minority-class detection substantially degrades under imbalanced conditions.

To assess generalisability beyond a single fraud detection domain, the study additionally evaluates on the Taiwan Default Payment dataset, a moderately imbalanced classification problem with mixed categorical features, incorporating SMOTE and TVAE as additional baselines alongside CTGAN and FinDiff.

The analysis reveals that class-ratio manipulation interacts with generative architectures in fundamentally different ways. Diffusion-based generators exhibit strong sensitivity to ratio forcing, leading to distributional distortion and monotonically degraded detection performance. In contrast, CTGAN often fails to enforce the requested minority ratios due to its conditional sampling mechanism, resulting in generated datasets that remain close to the natural class distribution. This effect paradoxically preserves downstream performance. Furthermore, a double-penalisation effect is identified in which applying both synthetic ratio forcing and classifier-level class reweighting using `scale_pos_weight` simultaneously causes catastrophic

performance collapse, demonstrating that these two imbalance-handling strategies are substitutes rather than complements.

To investigate these issues systematically, this study addresses the following research questions:

- **RQ1.** Can synthetic tabular data meaningfully improve or replace real data for fraud detection when evaluated using downstream classification performance?
- **RQ2.** How does explicit class-ratio manipulation during synthetic data generation affect data distribution and downstream detection performance under severe class imbalance?
- **RQ3.** Do different generative paradigms, namely GAN and diffusion, respond differently to class-ratio control in terms of distribution fidelity and downstream model performance?

2 Background and Related Work

2.1 Fraud Detection and Class Imbalance

Financial fraud detection is commonly formulated as a highly imbalanced binary classification problem, where fraudulent transactions represent only a small fraction of the total dataset. In real-world financial systems, the fraud ratio is often well below 1%, which causes standard machine learning models to become biased toward the majority class and results in poor detection of minority fraud cases.

Various approaches have been proposed to address this issue. Algorithm-level methods include cost-sensitive learning and class weighting, which increase the penalty associated with misclassifying minority instances. For example, gradient boosting frameworks such as XGBoost incorporate the parameter `scale_pos_weight` to adjust the relative importance of positive samples during training [4].

Data-level approaches attempt to rebalance the training distribution through resampling techniques. A widely used method is SMOTE, which generates synthetic minority samples through linear interpolation between existing minority observations [5]. However, SMOTE operates at the feature level without modelling the joint data distribution, and may struggle to capture complex feature dependencies present in high-dimensional financial transaction data. This limitation motivates the use of deep generative models as an alternative means of synthetic data generation. Importantly, Saito and Rehmsmeier [3] demonstrated that ROC-AUC can remain misleadingly high even when minority-class precision degrades, arguing that AUPRC is a more informative evaluation metric for imbalanced classification tasks.

2.2 Synthetic Data for Tabular Data

Recent advances in deep generative models have enabled new approaches for synthetic tabular data generation. Generative Adversarial Networks have been widely applied in this domain due to their ability to model complex data distributions. In particular, CTGAN was proposed to better handle mixed data types and imbalanced categorical variables commonly found in tabular datasets [1]. CTGAN employs a conditional vector alongside a mode-specific normalisation and rejection sampling mechanism to control the distribution of discrete columns during generation. Despite its effectiveness, GAN-based approaches can suffer from training instability and mode collapse, which may become more pronounced in highly imbalanced datasets.

More recently, diffusion-based generative models have emerged as a compelling alternative. Diffusion models generate data through an iterative denoising process that progressively transforms Gaussian noise into structured samples [6, 7]. FinDiff extended this framework specifically to financial tabular data, incorporating class-conditional label guidance at each denoising step to enable class-aware generation [2]. Extensions of diffusion models to general tabular settings, such as TabDDPM, have also demonstrated competitive performance in modelling complex tabular distributions [8]. This study selects CTGAN and FinDiff as the primary comparison pair for three reasons: both are identified as paradigm representatives in recent surveys of synthetic tabular generation [9]; diffusion-based tabular generators remain substantially less studied than GAN-based approaches under extreme class imbalance [10, 11]; and both provide fully open-source implementations that support the ratio-forcing experiments central to this study.

2.3 Evaluating Synthetic Data for Downstream Tasks

Prior research on synthetic tabular data primarily evaluates the quality of generated samples through statistical similarity between real and synthetic datasets. Common evaluation methods include feature-wise distribution comparisons, correlation preservation, and statistical distance metrics such as the Kolmogorov Smirnov statistic and Wasserstein distance. These metrics provide useful indicators of distributional fidelity and help assess whether the generative model captures the underlying structure of the data.

However, statistical similarity alone does not guarantee that synthetic data will be useful for downstream machine learning tasks. As demonstrated by Saito and Rehmsmeier [3], evaluation metrics must be chosen to reflect the actual prediction objective. In imbalanced settings, ROC-AUC can remain high even as the decision boundary for minority-class detection degrades, making Fraud F1 and AUPRC more appropriate primary metrics. Synthetic samples that appear statisti-

cally realistic may still fail to improve, or may actively harm, downstream model performance when used for classifier training.

2.4 Literature Review and Limitations

While previous studies have demonstrated strong distributional fidelity for synthetic tabular generators, several limitations remain when considering real-world fraud detection scenarios.

First, prior work on synthetic tabular generation has predominantly evaluated models under fixed class distributions that mirror the natural data ratio [12, 8]. As noted by D'souza et al. [10], class imbalance in the generative modelling process has not been studied as extensively as imbalanced classifier model training. How generators behave when class ratios are deliberately manipulated during sampling remains inadequately explored [13].

Second, the interaction between generative models and downstream classifier performance has received relatively limited attention. In fraud detection tasks, the ultimate objective is not only to reproduce the original data distribution but to improve predictive models' ability to detect rare events. Using downstream Fraud F1 and AUPRC as primary evaluation criteria, rather than relying solely on statistical distance metrics, directly addresses this gap.

Third, diffusion-based tabular generators are relatively recent, and their behaviour under extreme class imbalance remains less well characterised than their GAN-based counterparts [10, 11]. While recent work has begun to address this, for example by proposing imbalance-aware diffusion frameworks for financial data [11, 14], systematic comparisons between GAN and diffusion paradigms under controlled class-ratio manipulation settings remain lacking.

This study directly addresses these gaps by evaluating CTGAN and FinDiff under identical controlled conditions, using downstream fraud detection performance as the primary criterion, and explicitly testing the effect of class-ratio manipulation across a range of forced minority-class proportions.

3 Problem Definition

Fraud detection is formulated as a binary classification problem under severe class imbalance, where the minority class represents rare but operationally critical fraudulent events. The primary goal is to evaluate whether synthetic tabular data can meaningfully support fraud detection under this imbalance, and how explicit class-ratio manipulation during generation affects downstream classification performance.

Experiments are conducted primarily on the UCI Credit Card Fraud dataset (Kaggle: `mlg-ulb/creditcardfraud`), which contains 284,807 credit card transactions collected over two days in September 2013. All features V1–V28 are PCA-transformed to protect cardholder privacy; the remaining features are Time and Amount. The dataset exhibits extreme class imbalance at 0.173% fraud, yielding a class imbalance ratio of 577.9:1.

Table 1: Dataset Statistics

Property	DS1: Credit Card Fraud	DS2: Taiwan Default
Total samples	284,807	30,000
Minority class	492 (fraud)	6,636 (default)
Minority rate	0.173%	22.1%
Imbalance ratio	577.9:1	∼3.5:1
Feature type	Continuous (PCA)	Mixed (raw)
Features	30	23

DS2 (Taiwan Default Payment) is used in a generalisation study to assess whether findings on DS1 extend to a less severely imbalanced dataset with mixed categorical features. Results for DS2 are reported in Section 5.3.

4 Methodology

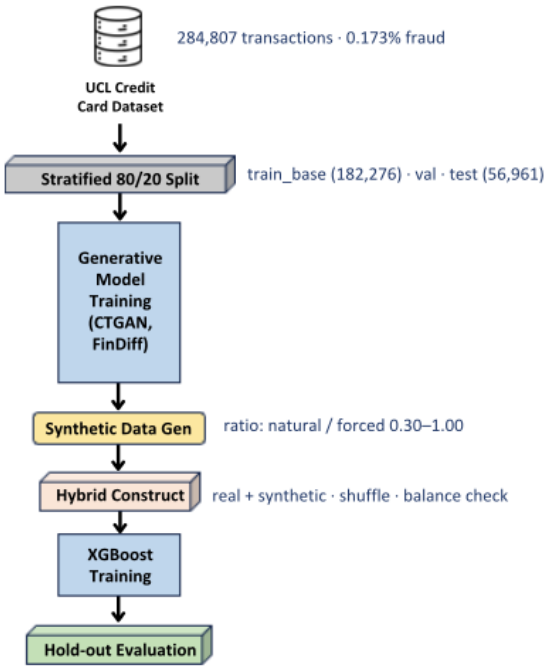


Figure 1: Experimental pipeline. The UCI Credit Card Fraud dataset is split once into `train_base`, validation, and a fixed hold-out test set. A generative model trained on `train_base` produces synthetic data under natural or forced class-ratio settings. The resulting hybrid or synthetic-only training set is then used to train a fixed XGBoost classifier, which is evaluated on real test data.

4.1 Experimental Pipeline

All experiments follow the pipeline shown in Figure 1. The dataset is split once using a stratified 80/20 hold-out with `random_state=42`, producing a fixed test set

used across all experiments. The generative model is trained exclusively on `train_base`. Synthetic data is generated at $N = 182{,}276$ in all conditions. For hybrid training, real and synthetic data are concatenated and shuffled. For synthetic-only training, `train_base` is replaced entirely by synthetic data. A `StandardScaler` is fitted on `train_base` only and applied to all other splits without refitting, preventing data leakage.

Table 2: Data Split

Split	N	Fraud cases
`train_base`	182,276	315
`val_df`	45,569	79
`test_df` (fixed)	56,961	98

4.2 Synthetic Data Generation Models

Two generative models are compared as the primary experimental subjects. SMOTE and TVAE are included as additional baselines in the generalisation study only (Section 5.3).

CTGAN [1] is a conditional GAN designed for tabular data. It uses a mode-specific normalisation and conditional vector to handle mixed data types. Ratio control is implemented via rejection sampling on the `condition_column`: the model generates batches and filters by class label until the target count is reached. Due to the GAN's inherent class bias (approximately 32% fraud under natural training), this mechanism frequently fails to deliver the requested ratio, as demonstrated in Section 5.2.

FinDiff [2] is a diffusion-based tabular generator that conditions generation on class labels injected at every denoising step. Ratio control is implemented by pre-assigning a label array of the desired class proportion before the reverse diffusion process begins, making ratio enforcement exact by construction.

4.3 Class Ratio Control

Ratio control is applied at generation time, not as a post-hoc resampling step. Five conditions are tested: natural, preserving the training distribution, and forced ratios of 0.30, 0.50, 0.70, and 1.00. The two models implement ratio control through structurally different mechanisms: rejection sampling in CTGAN versus label pre-assignment in FinDiff. This produces qualitatively different responses to ratio manipulation, as analysed in RQ2 and RQ3.

4.4 Downstream Classifier

All experiments use a fixed `XGBoost` classifier to isolate the effect of synthetic data quality from classifier choices. `XGBoost` is selected for its strong performance on tabular data [4, 15], its built-in support for class weighting via `scale_pos_weight`, and its sensitivity to distributional shifts in the training data, making it a suitable instrument for detecting changes in synthetic data quality.

Each experiment is run under both `spw` ON and `spw` OFF conditions to examine the interaction between synthetic class-ratio manipulation and classifier-level reweighting.

4.5 Evaluation Strategy

Table 3: `XGBoost` Fixed Hyperparameters

Parameter	Value
`n_estimators`	200
`max_depth`	6
`learning_rate`	0.1
`subsample`	0.8
`colsample_bytree`	0.8
`eval_metric`	`aucpr`
`random_state`	42
`scale_pos_weight` (ON)	577.9
`scale_pos_weight` (OFF)	1.0 (disabled)

Following Saito & Rehmsmeier[3], Fraud F1 and AUPRC are used as primary metrics, as ROC-AUC can remain misleadingly high even when minority-class detection degrades under severe imbalance. ROC-AUC is reported as a secondary reference metric. Distribution fidelity is assessed using the mean KS statistic and Wasserstein distance computed feature-wise between real and synthetic data. Privacy is assessed via nearest-neighbour distance and near-duplicate rate.

Decision threshold optimisation is applied using a validation set: `find_best_threshold()` sweeps thresholds in [0.001, 0.5] and selects the value maximising Fraud F1 on `val_df`. This threshold is then applied to the fixed test set.

Table 4: Evaluation Conditions

Dimension	Conditions
Training mode	Hybrid (real + syn) · Synthetic-only
`spw`	ON (577.9) · OFF
Ratio	natural · 0.30 · 0.50 · 0.70 · 1.00
Primary metrics	Fraud F1 · AUPRC
Secondary metrics	ROC-AUC · KS statistic · Wasserstein
Privacy metrics	NN distance · near-duplicate rate

4.6 Experimental Design

All experiments follow a one-at-a-time design, varying one condition at a time while holding others constant at their base values. The total experiment count is 50, structured as shown in Table 5.

Two design decisions were revised during development. First, initial plans for k-fold cross-validation were abandoned because training a generative model

Table 5: Experiment Structure

Group	Description	N
Baseline	Real-only, SMOTE; 2 datasets × 5 conditions	10
Ratio manipulation	5 ratios × 2 models × 2 `spw` conditions	20
CTGAN hyperparameter sweep	epochs, batch size, dim, `log_frequency`, discriminator steps	10
FinDiff hyperparameter sweep	epochs, batch size, MLP layers, diffusion steps, scaler	10
Total		**50**

per fold would have multiplied computational cost by 5×, making the full dataset infeasible with available resources. A fixed hold-out split was adopted instead. Second, `QuantileTransformer` preprocessing was replaced with `StandardScaler` for `FinDiff` after empirical testing showed improved downstream performance (Fraud F1: 0.851 → 0.874, AUPRC: 0.855 → 0.873); the mechanism underlying this difference is examined in Section 6.2.

5 Results

5.1 RQ1: Can Synthetic Data Improve or Replace Real Data?

Table 6 reports downstream performance for the best configurations of each training mode against the real-only baseline (5-fold CV, `spw` ON: Fraud F1 = 0.866, AUPRC = 0.8608).

Table 6: RQ1: Performance Comparison

Configuration	ROC-AUC	AUPRC	Fraud F1	Train data
Real-only (5-fold CV, spw ON)	0.9777	0.8608	0.866	Real only
FinDiff natural, hybrid (spw OFF)	0.9780	0.8730	0.860	Real + syn
CTGAN forced_0.30, hybrid (spw OFF)	0.9828	0.8412	0.832	Real + syn
FinDiff natural, syn-only (spw OFF)	0.9633	0.7247	0.774	Syn only
CTGAN forced_0.30, syn-only (spw OFF)	0.9780	0.7034	0.745	Syn only

Figure 2: Confusion matrices for the two best hybrid configurations: FinDiff natural hybrid (left) and CTGAN forced_0.30 hybrid (right), both `spw` OFF.

The confusion matrices in Figure 2 reveal the error profiles underlying these scores. FinDiff natural hybrid correctly identifies 83 of 98 fraud cases with only 12 false positives (Fraud F1 = 0.860, Legitimate F1 = 0.999). CTGAN forced_0.30 hybrid identifies 82 true positives with 17 false positives (Fraud F1 = 0.832, Legitimate F1 = 0.9997). Both configurations maintain high legitimate-class accuracy, confirming that synthetic augmentation does not disrupt majority-class performance. Under synthetic-only training, FinDiff natural achieves Fraud F1 = 0.774, indicating that full data replacement remains viable but incurs a non-trivial detection penalty. No near-duplicates were detected across any condition (NearDup rate = 0).

A consistent observation is that ROC-AUC remains stable across all configurations (0.963–0.983) even as Fraud F1 varies substantially. This motivates the use of Fraud F1 and AUPRC as primary metrics throughout this study [3].

5.2 RQ2: Effect of Ratio Forcing on Distribution and Performance

Table 7: FinDiff: Ratio Forcing Effects (hybrid, `spw` OFF)

Ratio	Fraud F1	KS Mean	Wasserstein	Syn. fraud rate
natural	0.860	0.016	68.0	0.173%
forced 0.30	0.463	0.122	548	30.0%
forced 0.50	0.337	0.203	939	50.0%
forced 0.70	0.263	0.286	1318	70.0%
forced 1.00	0.173	0.408	1,888	100.0%

FinDiff exhibits strict monotonic degradation across both distribution fidelity and downstream performance as the forced ratio increases, with no exceptions. KS rises from 0.016 to 0.408, and Fraud F1 collapses from 0.860 to 0.173, coinciding with the natural fraud rate and suggesting the classifier reduces to near-random minority-class prediction at the extreme. Unlike CTGAN, FinDiff enforces the requested ratio exactly through label pre-assignment before the reverse diffusion process, meaning the synthetic fraud rate matches the target precisely in all conditions. This makes the monotonic degradation a direct consequence of ratio forcing rather than a sampling artefact.

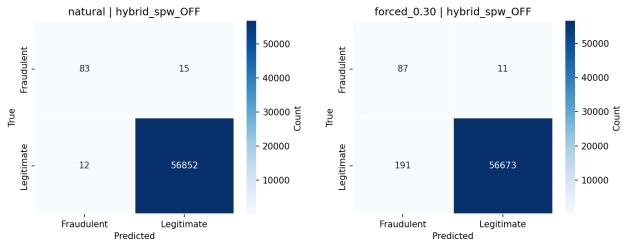


Figure 3: Confusion matrices for FinDiff hybrid: natural (left) and forced_0.30 (right), `spw` OFF.

The confusion matrices in Figure 3 make this collapse concrete. Under the natural ratio (synthetic fraud rate = 0.173%), the classifier produces 83 true positives and 12 false positives. Under forced_0.30 (synthetic fraud rate = 30%), true positives increase slightly to 87, but false positives explode to 191, a 16×

increase. The decision boundary has shifted: the classifier becomes aggressive but imprecise, flagging many legitimate transactions as fraud. This reflects a corrupted feature space in which synthetic fraud samples, generated at 30% prevalence from only 315 real fraud examples, occupy regions inconsistent with the true minority-class distribution.

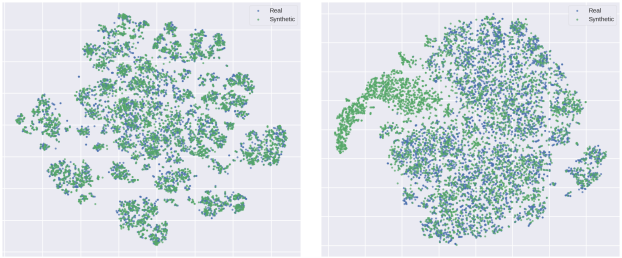


Figure 4: t-SNE visualisations of real data (blue) vs. synthetic data (green) for FinDiff: natural ratio (left) and forced_0.50 (right).

The t-SNE visualisations in Figure 4 provide a direct geometric explanation of the distribution shift induced by ratio forcing. Under the natural ratio, synthetic samples broadly overlap with the real data distribution across the full feature space, indicating that FinDiff has faithfully reproduced the overall data manifold. Under forced_0.50, where 50% of the 182,276 synthetic samples are labelled as fraud against only 315 real fraud cases in `train_base`, the synthetic distribution visibly diverges from the real distribution. Synthetic samples concentrate in regions of the feature space inconsistent with real data, and the overall overlap between real and synthetic deteriorates. This geometric divergence is directly reflected in the KS statistic rising from 0.016 to 0.203 between the natural and forced_0.50 conditions (Table 7), and explains the false positive explosion observed in Figure 3: the classifier learns a decision boundary shaped by an artificial synthetic distribution rather than the true data structure. This is attributed to label conditioning failure – FinDiff injects class labels at every denoising step, and forcing generation of 54,683 fraud samples (forced_0.30) or 91,138 fraud samples (forced_0.50) from a model trained on only 315 real fraud examples pushes the generative process far beyond the minority-class distribution it learned during training.

5.3 RQ3: GAN vs. Diffusion: Paradigm Differences

Table 8: CTGAN vs. FinDiff: KS and Fraud F1 by Ratio (hybrid, `spw` OFF)

Ratio	CTGAN KS	FinDiff KS	CTGAN F1	FinDiff F1	KS winner	F1 winner
natural	0.134	0.016	0.746	0.860	FinDiff	FinDiff
forced 0.30	0.132	0.122	0.832	0.463	≈ Tie	CTGAN
forced 0.50	0.165	0.203	0.783	0.337	CTGAN	CTGAN
forced 0.70	0.227	0.286	0.619	0.263	CTGAN	CTGAN
forced 1.00	0.378	0.408	0.644	0.173	CTGAN	CTGAN

Table 9: CTGAN: Requested vs. Delivered Synthetic Fraud Rate

Requested ratio	Delivered syn. fraud rate	Direction
natural	∼32.0%	+185× true rate
forced 0.30	∼11.0%	−19pp under-delivery
forced 0.50	∼53.6%	+3.6pp over-delivery
forced 0.70	∼96.5%	+26.5pp over-delivery
forced 1.00	∼100%	≈ on target

Under the natural ratio, FinDiff achieves substantially better distribution fidelity (KS 0.016 vs. 0.134) and downstream performance (Fraud F1 0.860 vs. 0.746).

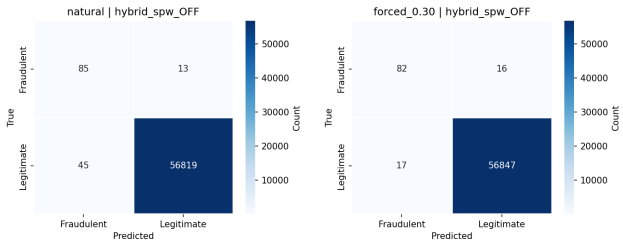


Figure 5: Confusion matrices for CTGAN hybrid: natural (left) and forced_0.30 (right), `spw` OFF.

Figure 5 (left) shows that CTGAN natural hybrid produces 85 true positives but 45 false positives – nearly 4× the false positive count of FinDiff natural (12). This is attributable to CTGAN's inherent class bias: even under natural sampling, the delivered synthetic fraud rate is approximately 32%, inflating the minority cluster and pushing the decision boundary outward.

Once ratio forcing is applied, the performance relationship inverts – but Table 9 reveals that CTGAN's robustness is a consequence of rejection sampling failure rather than architectural resilience. CTGAN filters generated batches by class label until the target count is reached, but because its internal distribution is biased toward ∼32% fraud, it consistently under-delivers below this point and over-delivers above it. Forced_0.30 delivers only 11% fraud; forced_0.70 delivers 96.5%. The confusion matrix for forced_0.30 (Figure 5, right) shows 82 true positives and 17 false positives – almost identical to FinDiff natural – precisely because the delivered rate of 11% is close to the natural training rate, leaving the hybrid distribution only modestly perturbed.

A final observation is that similar KS values at forced_0.30 (CTGAN: 0.132, FinDiff: 0.122) do not produce similar Fraud F1 scores (0.832 vs. 0.463). As shown in Table 9, CTGAN's KS at this setting reflects a delivered rate well below the requested 30%, close to natural, with distortion spread broadly at low intensity. FinDiff's KS reflects an exact rate of 30%, with distortion concentrated in the minority-class feature region. Identical KS statistics can therefore mask qualitatively different distributional failures, demonstrating that KS alone is insufficient to predict down-

stream utility.

6 Ablation Study

6.1 Double Penalisation: Forced Ratio and scale_pos_weight

Table 10: Double Penalisation: Fraud F1 under spw ON vs OFF

Model	Ratio	spw OFF	spw ON	Δ
FinDiff	natural	0.860	0.820	−4.7%
FinDiff	forced 0.30	0.463	0.025	−94.6%
FinDiff	forced 0.50	0.337	0.013	−96.1%
CTGAN	natural	0.746	0.103	−86.2%
CTGAN	forced 0.30	0.832	0.057	−93.1%
CTGAN	forced 0.50	0.783	0.028	−96.4%

Under the natural ratio, applying spw ON causes moderate reduction for FinDiff (−4.7%) but already severe collapse for CTGAN (−86.2%), consistent with CTGAN's more distorted synthetic distribution under natural conditions. Under any forced ratio, the combination of synthetic oversampling and spw ON causes catastrophic collapse across both models (85–98% reduction), as shown in Table 10. The confusion matrices under these collapsed conditions show near-constant positive prediction: the classifier flags the overwhelming majority of legitimate transactions as fraud, reducing precision to near zero. The mechanism is straightforward – synthetic ratio forcing already shifts the training class distribution toward greater minority-class presence, partially compensating for natural imbalance. Simultaneously applying spw ON instructs the classifier to further amplify minority-class weight, producing a doubly penalised training signal that overwhelms the decision boundary entirely. These two strategies are substitutes, not complements. Full confusion matrices for all conditions are provided in Appendix A.

6.2 Preprocessing: StandardScaler vs. QuantileTransformer

Table 11: FinDiff: Preprocessing Comparison (hybrid, natural, spw OFF)

Metric	StandardScaler	QuantileTransformer
Train loss	0.119	0.141
KS Mean	0.017	0.017
Fraud F1	0.874	0.851
AUPRC	0.873	0.855

Despite identical KS scores, StandardScaler yields consistently higher downstream performance. KS equivalence confirms that distributional fidelity of generated samples is unaffected by preprocessor choice – the performance gap originates entirely in the downstream classifier stage. QuantileTransformer maps all feature values to a uniform rank distribution, suppressing the relative magnitude of outliers that are diagnostically informative for fraud detection. StandardScaler preserves these extreme values, yielding lower training loss and higher Fraud F1. This result provides a further illustration that identical KS statistics can coexist with materially different downstream performance, reinforcing the case for task-based evaluation metrics.

6.3 Hyperparameter Sensitivity

Table 12: FinDiff Hyperparameter Sweep (hybrid, natural, spw OFF)

Parameter	Values tested	F1 range
Epochs	100 / 300 / 500	0.843–0.878
MLP layers	$[256^2]$ / $[512^3]$ / $[1024^4]$	0.865–0.877
Diffusion steps	200 / 500 / 800	0.868–0.878
Batch size	256 / 512 / 768	0.860–0.879

FinDiff Fraud F1 is stable across all hyperparameter configurations (spread: 0.036), confirming that architectural choices are not the primary sensitivity in this model. The dominant source of performance variation is class-ratio manipulation, not model architecture.

For CTGAN, a notable anomaly was observed during the hyperparameter sweep: setting log_frequency=False collapsed the delivered synthetic fraud rate to approximately 0.17%, matching the natural training rate, and yielded Fraud F1 = 0.874 – the highest result achieved by any CTGAN configuration across all 50 experiments. This mirrors FinDiff's natural-ratio performance and further confirms that natural distribution preservation is the dominant driver of downstream utility regardless of generative paradigm.

6.4 Generalisation Study: Taiwan Default Payment Dataset

Table 13: Generalisation Study: DS1 vs. DS2

Model	DS1 Fraud F1	vs Real	DS2 Default F1	vs Real
Real-only	0.860	baseline	0.443	baseline
SMOTE hybrid	0.905	+4.5%	0.445	+0.5%
FinDiff hybrid	0.863	+0.3%	0.467	+5.5%
TVAE hybrid	0.813	−5.4%	0.453	+2.3%
CTGAN hybrid	0.750	−12.8%	0.446	+0.7%

FinDiff is the only model to rank in the top two on both datasets. SMOTE performs strongly on DS1 – where PCA-transformed continuous features make linear interpolation between existing minority samples plausible – but offers negligible benefit on DS2 (+0.5%), where raw categorical features render interpolated values semantically unrealistic. FinDiff, by learning the full joint distribution through iterative denoising, remains effective across both structural settings. TVAE [16] and CTGAN underperform

the real-only baseline on DS1 (-5.4% and -12.8% respectively). CTGAN's underperformance is consistent with its higher KS statistic and lower distribution fidelity under natural conditions (Table 8).

7 Discussion

This study investigated three research questions concerning the utility of synthetic tabular data for fraud detection under severe class imbalance. The results yield several findings that extend beyond the specific models and dataset examined.

On RQ1: augmentation versus replacement. FinDiff natural hybrid (Fraud F1 = 0.860) approached the real-only baseline (0.866) while training on fewer real samples, demonstrating that diffusion-based augmentation can match the utility of real data without full replacement. Synthetic-only training remained viable under the natural ratio (F1 = 0.774) but fragile; any deviation from the natural class distribution led to rapid performance degradation. The practical implication is that synthetic data is best used as a complement to real data rather than a substitute, particularly under extreme imbalance where the minority class is represented by only a few hundred real examples.

On RQ2: the cost of ratio forcing. Forcing higher fraud ratios in FinDiff produced strict monotonic degradation in both KS statistic and Fraud F1, with no exceptions across five ratio settings. The t-SNE visualisations and confusion matrix analysis jointly confirm that this degradation is distributional in origin: synthetic fraud samples occupy incorrect regions of the feature space, corrupting the classifier's decision boundary and producing false positive explosion rather than detection improvement. Crucially, ROC-AUC remained stable (0.977–0.985) throughout this collapse, confirming that it is an unreliable primary metric for imbalanced augmentation studies and that Fraud F1 and AUPRC are necessary to detect real performance changes [3].

On RQ3: paradigm differences. The two models exhibit fundamentally different failure modes. FinDiff enforces requested ratios exactly through label pre-assignment, making it sensitive to any deviation from its training distribution. CTGAN's rejection sampling mechanism introduces a paradox: its failure to achieve target ratios at most settings inadvertently preserves distribution fidelity, but this robustness is an artefact of sampling failure rather than architectural strength. The finding that similar KS statistics at forced_0.30 (CTGAN: 0.132, FinDiff: 0.122) produced drastically different Fraud F1 scores (0.832 vs. 0.463) demonstrates that KS alone cannot capture the downstream consequences of distributional failures concentrated in the minority-class feature region.

On double penalisation. The combination of synthetic ratio forcing and classifier-level reweighting via `scale_pos_weight` produced catastrophic collapse (85–98% F1 reduction) across both models and all forced ratio settings. This is a practically important finding: practitioners commonly apply both synthetic oversampling and cost-sensitive learning simultaneously as complementary imbalance mitigation strategies. This study provides direct empirical evidence that under extreme imbalance, these strategies are substitutes rather than complements, and that their simultaneous application is reliably detrimental.

Limitations. Several limitations should be acknowledged. First, the real-only baseline used a larger training set (227,845 samples) than the hybrid configurations (182,276 real + 182,276 synthetic), meaning the hybrid advantage is partly offset by a size discrepancy; a size-controlled comparison was not conducted. Second, ratio forcing was tested on a single primary dataset; generalisability to other fraud distributions is not guaranteed, though the generalisation study on DS2 provides partial evidence. Third, the UCI dataset's PCA-anonymised features prevent domain-level validation of synthetic fraud patterns. Fourth, the central RQ3 conclusion rests on a methodologically important caveat: as shown in Table 9, CTGAN's rejection sampling mechanism fails to deliver the requested ratios in nearly every condition, meaning the two models are not compared under fully equivalent experimental conditions. CTGAN's apparent resilience is an artefact of implementation failure rather than architectural superiority; a valid paradigm-level comparison would require a CTGAN variant capable of exact ratio enforcement. Fifth, privacy evaluation was limited to nearest-neighbour distance and near-duplicate rate; membership inference attacks were not conducted.

8 Conclusion and Future Work

This study evaluated CTGAN and FinDiff as representative GAN-based and diffusion-based tabular generators for fraud detection under severe class imbalance, with a focus on the downstream consequences of explicit class-ratio manipulation during generation.

Four principal conclusions are drawn. First, synthetic augmentation under the natural class ratio can approach real-data performance: FinDiff natural hybrid achieves Fraud F1 = 0.860 against a real-only baseline of 0.866, while maintaining zero near-duplicate rate. Second, ratio forcing consistently degrades FinDiff performance through distribution shift in the minority-class feature region, with Fraud F1 collapsing from 0.860 to 0.173 as the forced fraud ratio increases from natural to 1.00; CTGAN shows a different response, but its apparent robustness reflects rejection sampling failure rather than architectural resilience. Third, ROC-AUC is an unreliable metric for synthetic augmentation studies under extreme imbalance, remaining stable at 0.977–0.985 while Fraud F1 collapses; Fraud F1 and AUPRC are the appropriate

primary criteria. Fourth, combining synthetic ratio forcing with classifier-level reweighting causes catastrophic performance collapse; these two imbalance-handling strategies must be treated as substitutes, not complements.

The generalisation study on the Taiwan Default Payment dataset further shows that FinDiff is the only model to rank in the top two across both datasets, suggesting that diffusion-based generation offers more consistent utility across varying imbalance levels and feature types than GAN-based alternatives.

Three directions for future work are identified. First, a size-controlled comparison between real-only and hybrid training would isolate the effect of synthetic augmentation from training set size differences. Second, stronger privacy evaluation through membership inference attacks would provide more rigorous privacy guarantees for synthetic financial data, which is important for regulatory compliance in real-world deployment. Third, extending the ratio-forcing protocol to newer generative frameworks, including LLM-based tabular generators such as GReaT and CLLM, would establish whether the sensitivity to ratio manipulation observed here is specific to diffusion and GAN architectures or a more general property of deep generative models applied to extreme minority-class settings.

Contributions and Use of AI

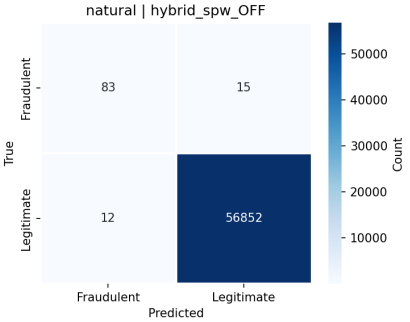
Implementation. CTGAN is implemented using the official SDV library (`sdv-dev/CTGAN`), extended with a `ratio` parameter to support class-ratio control at generation time. FinDiff is reproduced from the original authors' repository (`nussgen/FinDiff`) without modifications to the core architecture. No changes were made to the generative architectures of either model; the only addition was ratio control at the sampling stage.

Independent work. All research design decisions were made independently, including the formulation of research questions, experimental framework, dataset selection, and the design of class-ratio manipulation experiments. All experimental implementation was carried out independently, including the ratio-forcing pipeline, parameter exploration, and evaluation pipeline construction. AI assistance was used only for debugging specific coding errors during pipeline construction, strictly following the methodology described in the literature, and was not involved in designing, modifying, or interpreting any experimental logic. All result interpretation and analysis are my own.

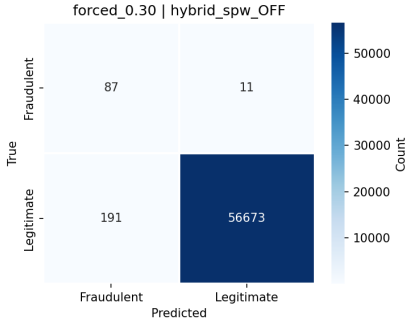
Use of AI assistance. AI tools were used in a limited and supplementary capacity. AI assistance was used to help interpret and summarise prior literature on diffusion models, CTGAN architecture, and evaluation metrics such as AUPRC and KS distance, and to help explain the structure of existing model implementations described in research papers.

Appendix A Confusion Matrices

A.1 FinDiff – Hybrid, spw OFF



natural



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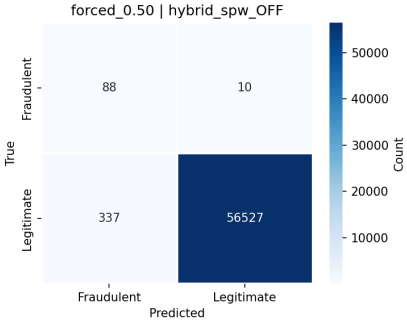

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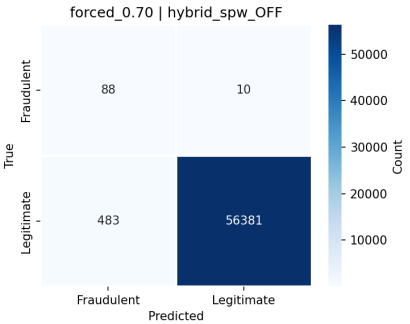
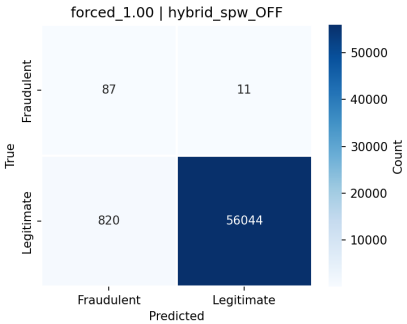
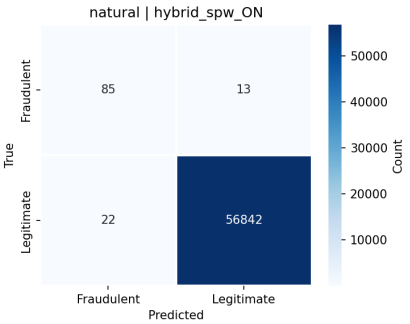


forced 0.70

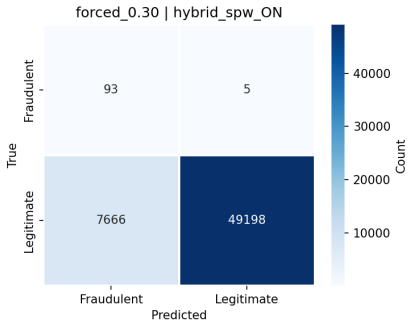


forced 1.00

A.2 FinDiff – Hybrid, spw ON



natural



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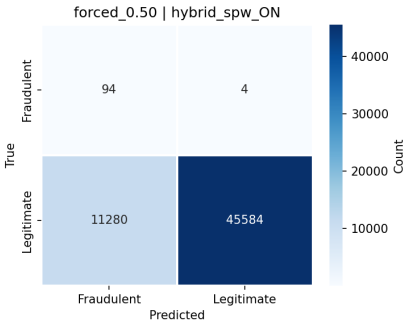

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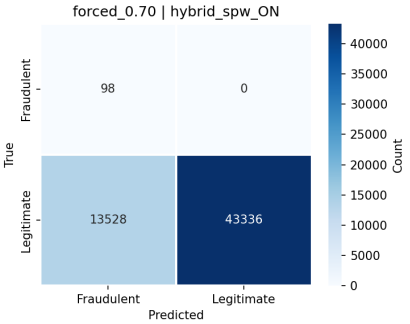

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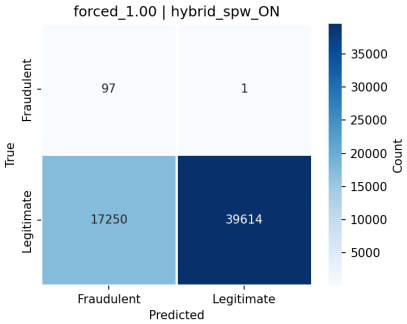

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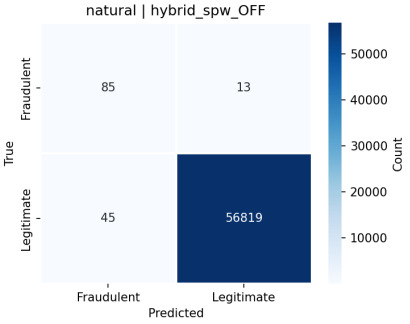
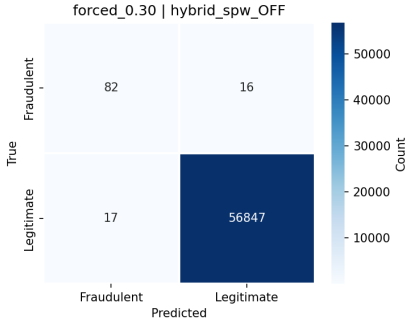
A.3 CTGAN – Hybrid, spw OFF



natural



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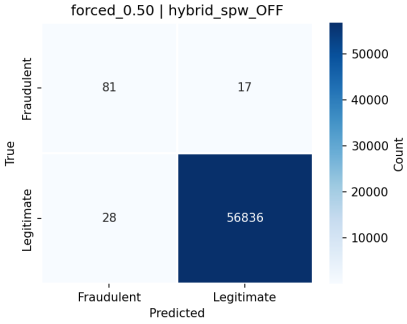

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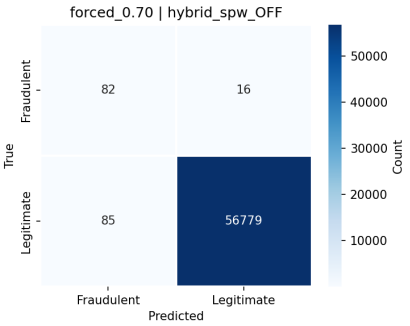

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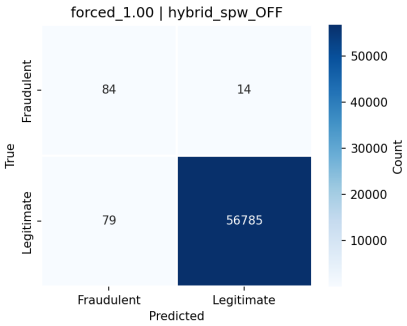

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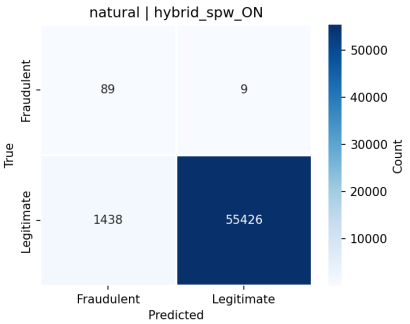
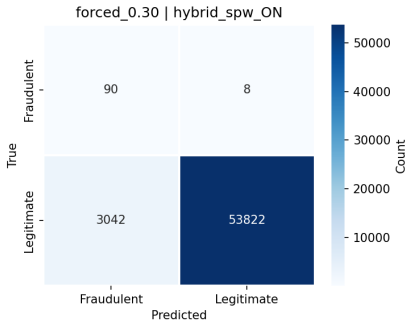
A.4 CTGAN – Hybrid, spw ON



natural



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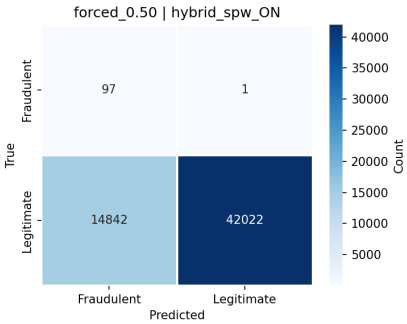

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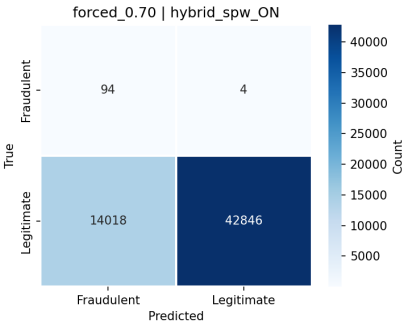

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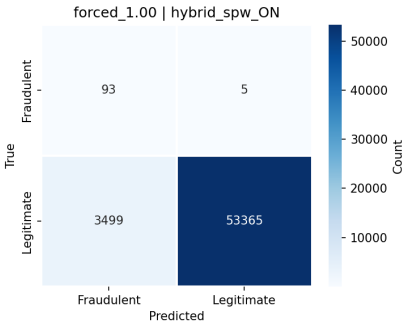

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